



FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES

SOFTWARE REQUIREMENT ANALYSIS (SRA)

SEMESTER	:	4
COURSE NAME	:	SOFTWARE ENGINEERING: THEORIES AND PRINCIPLES
LECTURER	:	
SYSTEM NAME	:	BOOKTRACK MANAGEMENT SYSTEM
PROGRAM NAME	:	BACHELOR OF COMPUTER SCIENCE (HONS.)
CONTACT PERSON	:	

STUDENT ID	STUDENT NAME	GROUP	ROLE
	NOR AFIQAH BINTI CIK AZAHARUDDIN		LEADER
	NUR AUNI INSYIRAH BINTI ZULKIPLI		SOFTWARE DESIGNER
	NURIN FARALIYANA BINTI MUHD YUSRI		SOFTWARE ENGINEER
	NUR AMALIN BINTI MOHD ANEEFUAZMI		SOFTWARE REQUIREMENT ANALYST

DATE OF SUBMISSION	27 MEI 2025 (before 6 pm)
ACTUAL DATE OF SUBMISSION	
NO OF DAYS LATE	

WE CERTIFY THAT THIS ASSIGNMENT IS OUR OWN ORIGINAL WORK AND MATERIALS HAVE NOT BEEN COPIED FROM ANY OTHER WORK (PUBLISHED OR OTHERWISE) WITHOUT ACKNOWLEDGEMENT IN THE TEXT. WE ARE FULLY AWARE OF THE PENALTY OF PLAGIARISM.

Table of Contents

1.	Organization chart	3
2.	Role and responsibility	4
3.	Introduction	5
4.	System overview	6
5.	Purpose	7
5.1	Scope	7
5.1.1.	Users	7
5.1.2.	Features	7
5.2.	Target Audience	8

1. Organization chart

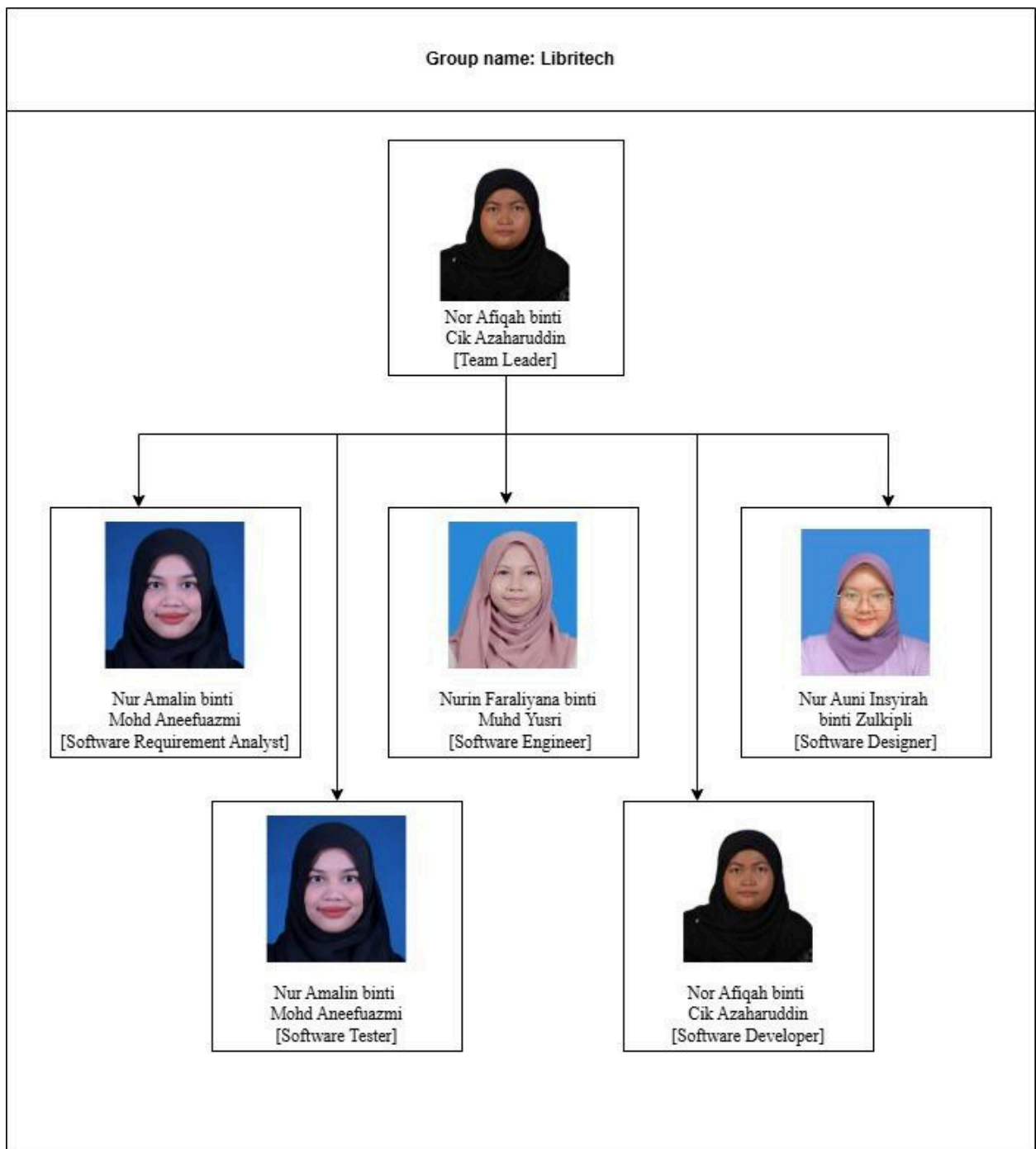


Figure 1-6 : Organization Chart

2. Role and responsibility

Table 1-1 : JobDescription

Name	RoleDescription
Nor Afiqah binti Cik Azaharuddin [Team Leader]	• Oversees the entire development process and ensures project milestones are met. • Coordinates team communication, assigns tasks, and facilitates problem-solving and decision-making.
Nur Amalin binti Mohd Aneefuazmi [Software Requirement Analyst]	• Gathers, analyzes, and documents functional and non-functional requirements for the BookTrack Management System. • Communicates with client to ensure their expectations are reflected in the system requirements.
Nur Auni Insyirah binti Zulkipli [Software Designer]	• Creates the architecture and design of the BookTrack Management System, including UI/UX layouts, data flow diagrams, and system models. • Ensures the design aligns with requirements and is scalable and maintainable.
Nur Amalin binti Mohd Aneefuazmi [Software Tester]	• Develops test cases and performs system, integration, and user acceptance testing to ensure the system meets quality standards. • Identifies bugs or issues and collaborates with developers to ensure they are resolved.
Nurin Faraliyana binti Muhd Yusri [Software Developer]	• Implements the system functionalities including book search, genre sorting, reading tracking, and user interaction features. • Writes and maintains clean, efficient code based on design specifications.
Nor Afiqah binti Cik Azaharuddin [Software Engineer]	• Integrates all system components, ensuring compatibility between front-end and back-end modules. • Optimizes performance and ensures the system is secure, robust, and ready for deployment.

3. Introduction

As digital technologies continue to evolve, people are now relying on information systems to optimize data management, personal organization and even professional routines. These systems provide users with controls for structured storage, segmentation, classification, indexing, retrieval and information sharing, which leads to effectiveness, precision and ease of use. Such systems mitigate data fragmentation, enable effective data retrieval and ensure consistent user experience with updated information through data centralization. This approach is particularly valuable in personal information management, where tracking activities such as reading habits and organizing digital book collections is essential for a personalized and streamlined experience.

Book lovers often face challenges in managing their reading habits effectively. While many digital platforms are available to catalog books, but most of them lack advanced functionalities like tracking reading progress, organizing eBook collections or engaging with a reading community. As users switch between apps and devices, their collections can become disorganized, leading to repeated purchases, forgotten titles or unfinished books. Without tools for motivation or interaction it will reduce user engagement and makes it difficult to maintain consistent reading behavior.

This project is intended to create an integrated web based BookTrack Management System which acts as a focal point for users to organize their individual libraries. Users can execute searches and add eBooks to their personalized lists, enter new titles which are not part of the existing inventory, as well as keep a wish list for books they intend to purchase. Furthermore, users can rate and leave reviews as well as comments for the books they have read, adding value to personal reflection as well as interaction within the community. This system solves existing problems by combining social and motivational tools with book tracking features.

Despite its advantages, several limitations must be considered when developing this system. Managing a vast and diverse collection of digital books requires significant time for data collection, validation and organization. Moreover, the system may not initially accommodate all genres or formats of digital books. However, the BookTrack Management System represents a meaningful step toward helping readers better manage their digital libraries, stay motivated and connect with a broader community of book lovers.

4. System overview

The BookTrack Management System is an interactive, web-based platform designed to help individual readers efficiently manage their digital book collections and reading activities. It combines features of personal library organization, real-time reading progress tracking, community engagement, and purchase management to offer a comprehensive solution for modern readers.

The system addresses the inefficiencies of traditional and fragmented book tracking methods such as physical journals, spreadsheets, or limited-function mobile apps by centralizing book information, automating genre categorization, and enabling personalized features like wishlists and reading analytics.

Key Functionalities:

- **User Profile Management:** Users can register, create personalized profiles, and track their reading goals and history.
- **Digital Book Catalog:** Supports both manual and automated book entry, allowing users to build and manage custom shelves such as "Currently Reading", "Owned Books".
- **Reading Progress Tracker:** Offers visual indicators of progress (pages or percentage), goal setting, and timeline views to monitor habits over time.
- **Genre-Based Organization:** Enables efficient sorting and filtering of book collections based on genres, improving the browsing experience.
- **Wishlists and Purchase Links:** Users can create wishlists with smart purchase suggestions from external platforms such as Amazon or Shopee.
- **Community Features:** Includes book reviews, ratings, and user comments to foster engagement and promote quality content.
- **Search & Explore:** Advanced search tools with filters for title, author, genre, and user reviews.

5. Purpose

5.1 Scope

5.1.1. Users

The BookTrack Management System is intended for a wide range of individual users who require a centralized and efficient platform to manage their reading activities. The primary users include casual readers who seek a simple way to organize their personal book collections, as well as students who wish to track both academic and leisure reading. Educators are also part of the target user group, using the system to observe and support students' reading progress through analytics and activity monitoring.

In addition, the system is designed to support members of book clubs and independent reading communities. These users can utilize the platform to coordinate group reading goals, share reviews, and interact socially through comments and recommendations. The system emphasizes personal ownership and engagement, distinguishing it from institutional library systems, which focus on administrative control and lending functionality. As such, the platform is best suited for individual or community-based use, rather than formal educational institutions or public libraries.

5.1.2. Features

The BookTrack Management System offers a comprehensive set of features that enhance the personal reading experience. Users can create and manage personalized profiles, including details such as reading history, preferred genres, and reading goals. Social features allow users to follow others and observe their reading activities, fostering community interaction.

The system provides the ability to create a personal digital book catalog. Users can add books manually or import them from external sources such as Open Library and Google Books. Books can be organized into custom shelves, including categories such as “Currently Reading,” “Want to Read,” and “Owned Books.” Each book entry contains metadata such as the author’s name, genre, a brief description, and a cover image.

One of the core functionalities is real-time reading progress tracking. Users can log their reading progress by either page number or percentage and visualize

their progress through timelines. The system also allows for the setting of personal reading goals and sends reminders to help users stay on track.

To enhance organization and browsing, the system includes genre sorting features that allow users to categorize their collections automatically or manually based on literary genre. This feature supports easier discovery and a more personalized reading experience.

A wishlist function is also provided, enabling users to keep track of books they intend to acquire. The system enhances this feature by suggesting external purchase links from platforms such as Amazon, Shopee, and TikTok Shop, facilitating convenient access to books not currently in the user's collection.

Community features include the ability to leave reviews, provide ratings, and write comments on books. The system also compiles this data to generate trending book lists and tailored reading recommendations.

Finally, an advanced search engine allows users to locate books based on title, author, genre, or user reviews. This search functionality is supported by recommendation algorithms that analyze individual reading behavior to suggest books that match the user's interests.

Despite its robust feature set, the system does not support audiobooks, video content, or digital lending. Furthermore, it depends on third-party APIs for some book data and purchase links, which may introduce variability in data accuracy and availability. User-generated content will require moderation to maintain data integrity and prevent duplication.

5.2. Target Audience

The BookTrack Management System is designed to cater to a diverse audience of readers who seek a more structured, engaging and interactive approach to managing their reading activities. Specifically, it addresses the needs of users across different reading backgrounds by offering features that enhance organization, motivation and social interaction.

To begin with, casual readers who enjoy reading in their free time but lack an organized way to track progress or manage their collections will benefit from features

like customizable shelves, real-time progress tracking, and automatic genre classification. As a result, these tools help transform a relaxing hobby into a more satisfying and rewarding experience, allowing users to revisit their reading history and discover new books that match their interests.

In the academic context, students who often manage multiple readings simultaneously can use the platform to keep track of required textbooks, reference materials, and course-related readings. With this in mind, the ability to update reading progress and set reading goals allows students to plan their study time more efficiently and ensure that no assignments or readings are overlooked throughout the academic term.

Furthermore, book clubs will find value in the system's ability to facilitate shared book lists, assign group reading tasks, and track collective progress. Through features such as reviews, comments, and ratings, the platform also encourages group interaction, allowing members to share opinions and maintain meaningful discussions about the books they read together.

In conclusion, this broad target audience reflects the system's versatility and its ability to support both individual and group reading experiences. By addressing the specific needs of each group, the BookTrack Management System positions itself as a valuable tool in both casual and academic reading contexts.